

Week 3: Beyond Triangles

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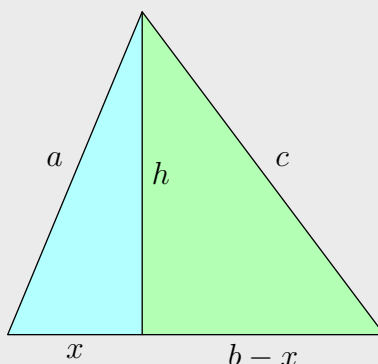
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※ Area of a Triangle

You may be familiar with the classic $\frac{1}{2}bh$ if you were given the base and the height of a triangle. Suppose that, instead, you were only given the side lengths of the triangle.

Consider the following triangle with side lengths a , b , and c . Use Pythagorean Theorem to find the area.



Solution. By looking at the two right-angled triangles (the blue and the green), we can establish two relationships:

- For the left (blue) triangle: $x^2 + h^2 = a^2$
- For the right (green) triangle: $(b-x)^2 + h^2 = c^2$

Setting them equal to each other:

$$a^2 - x^2 = c^2 - (b-x)^2$$

If we expand $(b-x)^2$, we get $b^2 - 2bx + x^2$. Substituting this back in:

$$a^2 - x^2 = c^2 - (b^2 - 2bx + x^2) \implies a^2 - x^2 = c^2 - b^2 + 2bx - x^2.$$

Notice that the $-x^2$ terms on both sides cancel out, leaving us with a linear equation for x :

$$a^2 = c^2 - b^2 + 2bx \implies x = \frac{a^2 + b^2 - c^2}{2b}.$$

Now we substitute this expression for x back into our first equation ($h^2 = a^2 - x^2$):

$$h^2 = a^2 - \left(\frac{a^2 + b^2 - c^2}{2b} \right)^2$$

To make this manageable, we use the difference of squares identity, $u^2 - v^2 = (u-v)(u+v)$:

$$h^2 = \left(a - \frac{a^2 + b^2 - c^2}{2b} \right) \left(a + \frac{a^2 + b^2 - c^2}{2b} \right)$$

Combining the terms over a common denominator ($2b$):

$$h^2 = \left(\frac{2ab - a^2 - b^2 + c^2}{2b} \right) \left(\frac{2ab + a^2 + b^2 - c^2}{2b} \right)$$

Look closely at the numerators. We can group a^2 , $2ab$, and b^2 into perfect square trinomials:

- The first part: $c^2 - (a^2 - 2ab + b^2) = c^2 - (a - b)^2$
- The second part: $(a^2 + 2ab + b^2) - c^2 = (a + b)^2 - c^2$

Now our equation for h^2 looks like this:

$$h^2 = \frac{[c^2 - (a - b)^2] \cdot [(a + b)^2 - c^2]}{4b^2}.$$

Applying the difference of squares one more time to each bracketed term:

$$h^2 = \frac{(c - a + b)(c + a - b)(a + b - c)(a + b + c)}{4b^2}.$$

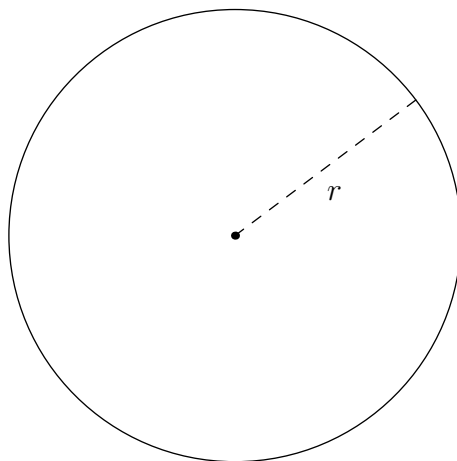
Taking the square root gives us the height h . Since the area is $\frac{1}{2}bh$, we multiply the height by $\frac{b}{2}$:

$$\text{Area} = \frac{b}{2} \sqrt{\frac{(a + b + c)(a + b - c)(a + c - b)(b + c - a)}{4b^2}} = \sqrt{s(s - a)(s - b)(s - c)}$$

where s is the semi-perimeter $s = \frac{a+b+c}{2}$.

✂ Circles

Given a point P and some constant r . A circle is the set of points which are a distance of r from P . A circle can be drawn as:

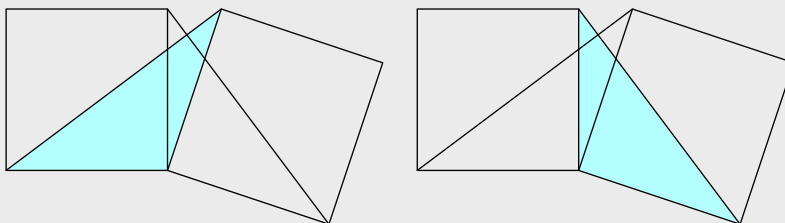


Some common facts are: the perimeter (or circumference) of the circle is $2\pi r$ while the area is πr^2 .

My biggest tip to you is: When you see tangencies, centers, points on the circumference of the circle, connect them *reasonably*. In general, you won't find much with circles. A lot of formulas are concerned with lengths of figures. So, draw lengths and you will get more out of it. And finally: just because a point is not there doesn't mean you cannot create it!

※ Congruency

Two squares share a vertex, as shown below. What is the relationship between the shaded triangles?



Solution. Let's look at the sides that meet at the shared vertex (where the two squares touch):

- The Large Square Side: The first triangle uses a side of the large square, and the second triangle also uses a side of the large square. Since all sides of a square are equal, these two lengths are identical.
- The Small Square Side: Similarly, both triangles use one side from the smaller, tilted square. Since all sides of that square are equal, these two lengths are also identical.

This is the clever part. Let's look at the "gap" between the squares and call that angle β .

- Triangle 1 (Left) has an angle at the shared vertex equal to $90^\circ + \beta$.
- Triangle 2 (Right) also has an angle at the shared vertex. If you rotate the large square 90° to match the orientation of the other, you'll see that the angle formed between the two squares' sides is also $90^\circ + \beta$.

Because both triangles share:

- A side of length s_1
- A side of length s_2
- An identical angle of $(90^\circ + \beta)$ between those sides

We can conclude that the triangles are congruent. If you were to cut one out, you could rotate it and overlay it perfectly on top of the other.

In general, there are four ways we can prove congruency:

- SSS: Three sides are equal.
- SAS: Two sides and the included angle are equal (used in the "Shared Vertex Squares" problem). Order matters.
- ASA / AAS: Two angles and one side are equal. Order matters.
- HL (Hypotenuse-Leg): Specifically for right triangles, if the hypotenuse and one leg match, the triangles are congruent.

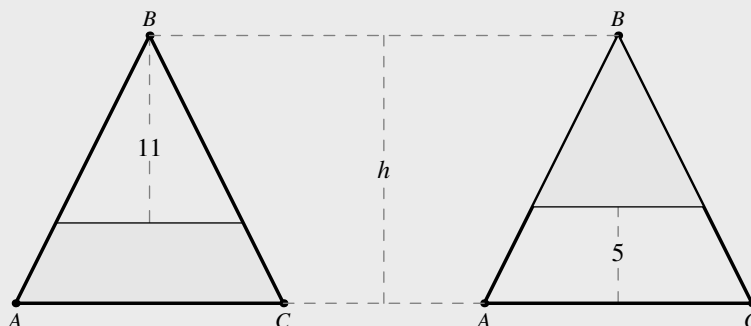
※ Similarity

Similarity follows a *similar* method. Note we remove ASA/AAS by replacing it with AA.

- SSS: Three sides have equal ratios. In other words, the sides of one triangle is a, b, c while the other is ak, bk, ck .
- SAS: Two sides have equal ratios and the included angle are equal (similar to what is used in the "Shared Vertex Squares" problem). Order matters.
- HL (Hypotenuse-Leg): Specifically for right triangles, if the hypotenuse and one leg have equal ratios, the triangles are similar.
- (NEW!) AA Similarity: If two angles match, the third must also match, making the triangles similar.

If the ratio between the sides of two similar triangles is k , then the ratio of their areas is k^2 . To see why, since $\text{Area} = \frac{1}{2}bh$, if you multiply the base by k and the height by k , the new area becomes $\frac{1}{2}(bk)(hk) = k^2(\frac{1}{2}bh)$.

Isosceles $\triangle ABC$ has equal side lengths AB and BC . In the figure below, segments are drawn parallel to $\overline{AC}$ so that the shaded portions of $\triangle ABC$ have the same area. The heights of the two unshaded portions are 11 and 5 units, respectively. What is the height of h of $\triangle ABC$?



Because the segments are parallel to the base, any triangle formed at the top of $\triangle ABC$ is **similar** to the large triangle. We use the property that if the ratio of the heights of two similar triangles is k , the ratio of their areas is k^2 . Let A_{total} be the area of $\triangle ABC$ and h be its total height.

From the left figure, the unshaded top triangle has height 11. Its area is:

$$\text{Area}_{\text{unshaded}} = \left(\frac{11}{h}\right)^2 A_{total}$$

The shaded region X is the total area minus this top triangle:

$$X = A_{total} - \left(\frac{11}{h}\right)^2 A_{total} = A_{total} \left(1 - \frac{121}{h^2}\right)$$

From the right figure, the unshaded portion at the bottom has height 5, meaning the shaded triangle at the top has a height of $(h - 5)$. Its area X is:

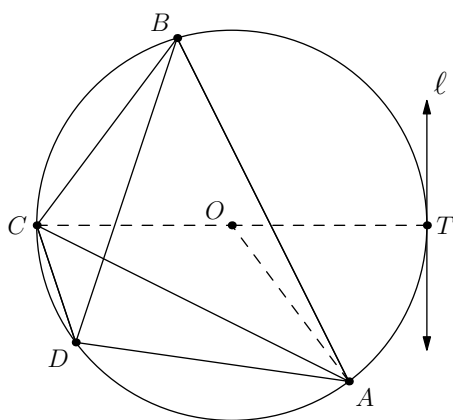
$$X = \left(\frac{h-5}{h}\right)^2 A_{total} = A_{total} \left(\frac{(h-5)^2}{h^2}\right)$$

Since both expressions represent the same area X , we set them equal:

$$\begin{aligned}
 1 - \frac{121}{h^2} &= \frac{(h-5)^2}{h^2} \\
 \Rightarrow h^2 - 121 &= (h-5)^2 \\
 \Rightarrow h^2 - 121 &= h^2 - 10h + 25 \\
 \Rightarrow -121 &= -10h + 25 \Rightarrow h = 14.6
 \end{aligned}$$

The height h of triangle ABC is 14.6 units.

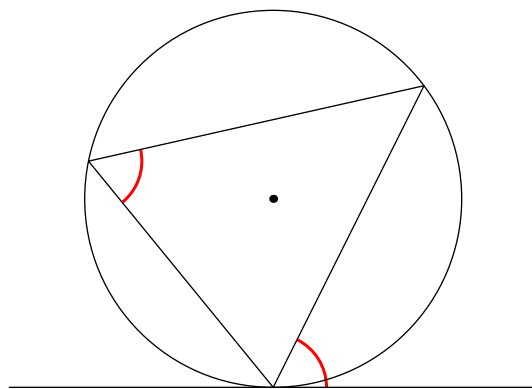
✂ A few angle properties of circles



1. Let A , B , and C be points on the circumference of a circle centered at O . Then, $2 \cdot \angle ABC = \angle AOC$.
2. Let A , B , C , and D be points on the circumference of a circle (in that order). Then, $\angle ABC + \angle CDA = 180^\circ$.
In fact, we say $ABCD$ is a *cyclic quadrilateral* if $\angle ABC + \angle CDA = 180^\circ$. This is because the opposite is true: if $\angle ABC + \angle CDA = 180^\circ$ then we can circumscribe a circle around $ABCD$.
3. Let A , B , C , and D be points on the circumference of a circle (in that order). Then, $\angle ACB \cong \angle ADB$.
4. Let T be a point on the circumference of a circle. If a line ℓ is tangent to T , then it intersects the circle at a single point being T . In particular, OT is perpendicular to ℓ !

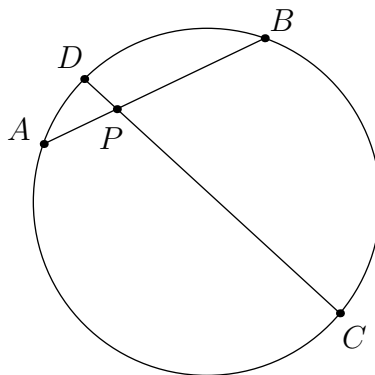
✂ Practice Problems

1. Show the two marked angles are equal.

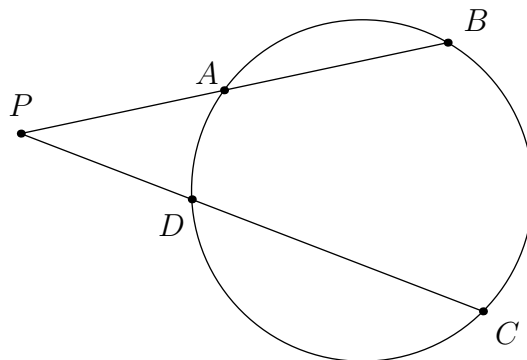


2. Prove *Power of the Point*. (tips: try angle chasing and look for similar triangles!)

- (a) If two chords AB and CD intersect at a point P within a circle, then $AP \cdot BP = CP \cdot DP$.

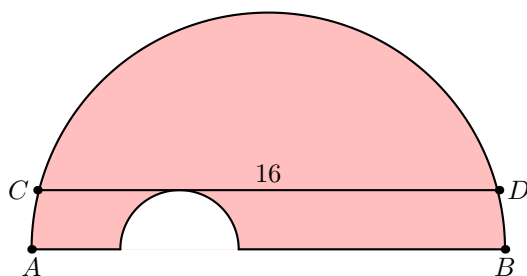


- (b) Given lines BP and CP originate from two unique points on the circumference of a circle (B and C), intersect each other at point P , outside the circle, and re-intersect the circle at points A and D respectively, then $PA \cdot PB = PD \cdot PC$.



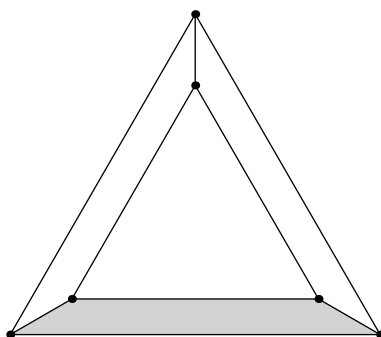
- (c) What do you think would happen in (b) if one of the lines were tangent to the circle? Can you prove it?

3. A semicircle has diameter $\overline{AB}$ and chord $\overline{CD}$ of length 16 parallel to $\overline{AB}$. A smaller semicircle with diameter on $\overline{AB}$ and tangent to $\overline{CD}$ is cut from the larger semicircle, as shown below.

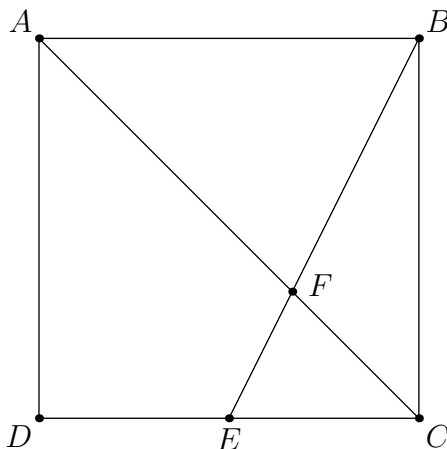


What is the area of the resulting figure, shown shaded?

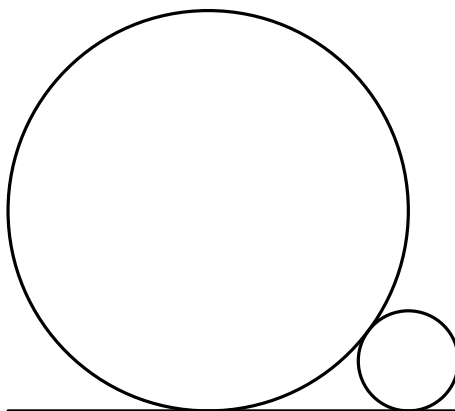
4. An equilateral triangle is placed inside a larger equilateral triangle so that the region between them can be divided into three congruent trapezoids, as shown below. The side length of the inner triangle is $\frac{2}{3}$ the side length of the larger triangle. What is the ratio of the area of one trapezoid to the area of the inner triangle?



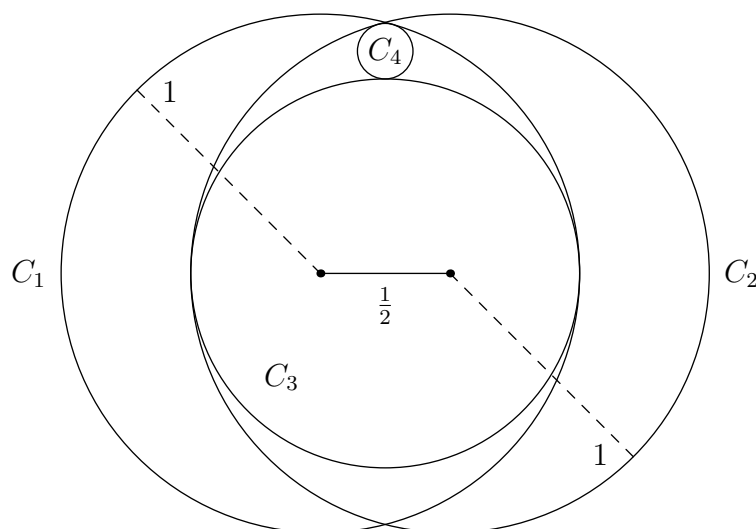
5. Point E is the midpoint of side $\overline{CD}$ in square $ABCD$, and $\overline{BE}$ meets diagonal $\overline{AC}$ at F . The area of quadrilateral $AFED$ is 45. What is the area of $ABCD$?



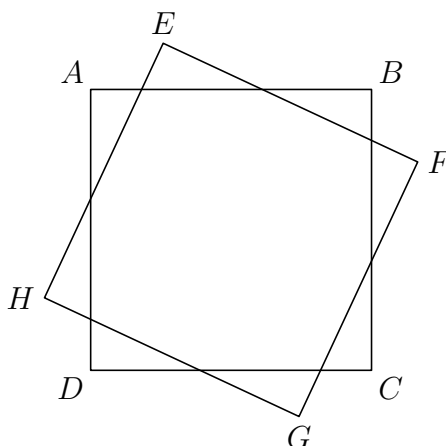
6. Two straight pipes (circular cylinders), with radii 1 and $\frac{1}{4}$, lie parallel and in contact on a flat floor. The figure below shows a head-on view. What is the sum of the possible radii of a third parallel pipe lying on the same floor and in contact with both?



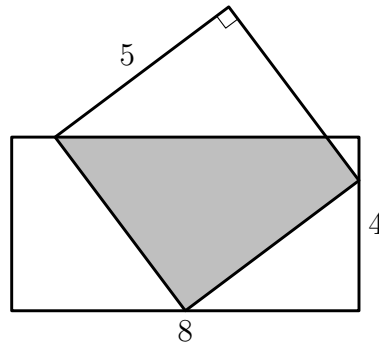
7. Circle C_1 and C_2 each have radius 1, and the distance between their centers is $\frac{1}{2}$. Circle C_3 is the largest circle internally tangent to both C_1 and C_2 . Circle C_4 is internally tangent to both C_1 and C_2 and externally tangent to C_3 . What is the radius of C_4 ?



8. Square $ABCD$ is rotated 20° clockwise about its center to obtain square $EFGH$, as shown below. What is the degree measure of $\angle EAB$?



9. The diagram below shows a rectangle with side lengths 4 and 8 and a square with side length 5. Three vertices of the square lie on three different sides of the rectangle, as shown. What is the area of the region inside both the square and the rectangle?



10. In the right triangle ABC , $AC = 12$, $BC = 5$, and angle C is a right angle. A semicircle is inscribed in the triangle as shown. What is the radius of the semicircle?

